

LESSON 9.4

FINDING AND INTERPRETING MEDIAN, MODE, AND RANGE



LESSON INTRODUCTION

MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode, and range.

LEARNING TARGETS

- I can find the mean, median, mode, and range of a given data set.
- I can interpret the median, mode, and range of a given data set.

BENCHMARK COHERENCE

BUILDING ON

MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.

WORKING TOWARD

MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.

ESSENTIAL QUESTIONS

- How can you find and interpret the mean, median, mode, and range of a given data set?
- What does the mean/median/mode/range of a data set tell you?
- How does context impact the interpretation of mean/median/mode/range of a data set?

LESSON NARRATIVE

In this lesson, students will explore and find the mean, median, mode, and range of a given data set. Students will interpret their findings to describe the data sets. Students will work with partners to explore the concepts of mean, median, and mode.

COMPONENT SUMMARY

9.4.1 WARM-UP (~5 MIN)

Notice and wonder about given means and medians

9.4.2 GUIDED INSTRUCTION (10 MIN)

Find mean, median, and mode using definitions

9.4.3 TRY IT (5 MIN)

Practice finding mean and median

9.4.4 EXPLORATION (10 MIN)

Practice finding mode and range

9.4.5 COLLABORATION (10-15 MIN)

Work with a partner to find mean, median, mode, and range of various data sets

9.4.6 WRAP-UP (~5 MIN)

Reflect on understanding of mean, median, mode, and range

RESOURCES

GLOSSARY

Key Terms:

- Mean
- Median
- Mode
- Range
- Measures of variability

Additional Terms: N/A

MATERIALS

- (Optional) Calculator
- (Optional) Computer

ADDITIONAL PREPARATION

- Consider predetermining partners before the lesson to help with matching students appropriately.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse mean and median or mode and median.
- Students may think the mean or median must be one of the values in the data set.
- Students may try to find the median before ordering the data set.

THINGS TO CONSIDER

- Students may feel overwhelmed with finding four different measurements. Consider providing graphic organizers or chunking information.
- Encourage students to write out all of the numbers in a data set in order from least to greatest every time they deal with the set to make sure they do not miss a number.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

The 9.4.1 Warm-Up provides an opportunity for students to engage in a Notice and Wonder routine to analyze data presented in a histogram and begin to consider the differences between the measures of center, mean and median. Use this as an opportunity to pique students' curiosity about how one measure of center may best describe data in different circumstances.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

9.4.5 Collaboration provides an opportunity for students to engage in project-based learning to connect the material from the lesson to their own experiences. Consider allowing students to find the heights of boys and girls in their own class and see how their data compares to the sample class. This will foster investment in the content and deepen understanding of the learning goals.

DIFFERENTIATE INSTRUCTION

This lesson is scaffolded throughout to provide multiple entry points for all learners. Struggling students will benefit from the additional support of multiple choice and word banks to accompany fill-in-the-blank questions. Discussion-based activities like the 9.4.5 Collaboration provide auditory entry points. Encourage students to create number lines or other graphic displays to provide additional visual entry points to synthesize and make sense of the data presented in the tables.

9.4.1 WARM-UP: NOTICE AND WONDER

Component Overview: Students record what they notice and wonder about a graph displaying mean and median income of several countries.

TEACHER MOVES

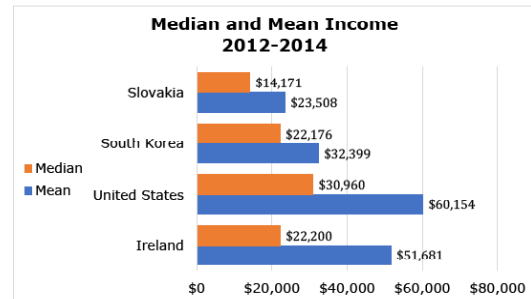
Notice and Wonder

Give students a minute or two to record what they notice and wonder about the graph. Circulate as students record their answers, look for students noticing the difference between the mean and median. Have a few students share out their notices and wonderings and focus attention on the large difference between mean and median in some countries and why that might be.



9.4.1 Warm-Up: Notice and Wonder

The diagram below shows bar charts for the median and mean income between 2012 and 2014 in selected countries. The income is given in U.S. dollars (\$).



1. What do you notice about the values for median and mean income?

Answers may vary. Mean is larger than median for all countries. Ireland and South Korea have similar median income, but very different mean income.

2. What do you wonder about the values for median and mean income?

Answers will vary. How is median calculated? How are median and mean different? How are they similar? When would you want to use mean? When would you want to use median?



9.4.2 Guided Instruction: Mean, Median and Mode

1. For each definition, fill in the blank with one of the words from the bank.

median	mode
mean	range



The mean is the arithmetic average of a set of numbers. It is a measure of central tendency.



The range is the difference between the highest data value and the lowest data value in a data set. It is a measure of variation.



The mode is the value found most often in a set of numbers. There may be no, one, or more than one mode in the data set. It is a measure of central tendency.



The median is the middle of an ordered list of the values. If the list has an odd number of values, it is the middle value of that list. If the list has an even number of values, it is the average of the two middle values. It is a measure of central tendency.

9.4.2 GUIDED INSTRUCTION: MEAN, MEDIAN, AND MODE

Component Overview: In this Guided Instruction, students begin by filling in the definitions using terms from a word bank. Go through the definitions with the class to ensure students understand what each term means before moving on. Students then find the mean, median, and mode of a data set and describe the steps to find each of them.

TEACHER MOVES

Fill in the blanks for the glossary terms with the class. Begin by reading a definition to the class. Then have students talk with a partner to find and define the term in their own words. Review the correct answers as a class.

DIFFERENTIATE INSTRUCTION

Allowing students to discuss each glossary term with a partner provides an auditory entry point to engage learners. Consider providing visual entry points, such as a histogram and data set with each term labeled, to engage visual learners.

The number of pets ten students in Tyler's class have is reflected in the table.

Number of Pets									
1	0	2	1	10	0	2	0	8	12

- Describe steps to find the median.
Order the numbers from least to greatest. Find the middle number. If there is an even number of entries in the data set, find the average of the two middle numbers.
- The median number of pets for the ten students is 1.5.
- The mode of the data set is 0.
- Write an explanation of what the mode of the data set means for the context.
Answers will vary. The most common response is zero pets.
- Describe the steps needed to find the average number of pets owned by the ten students.
Find the sum of all the values in the data set. Then divide the sum by the number of values in the data set, 10.
- The mean number of pets for the ten students is 3.6.
- Complete the statement about the mean of the data set.
"Each student in Tyler's class has about 3 or 4 pets."

ENRICHMENT

Students may struggle to describe the steps to find the median. Provide support through questioning:

- What is the first thing you need to do to the data set to begin to find the median? Why?
- After the numbers are in order, how can you find the median? Is the number of data points significant? Why?

9.4.3 TRY IT: MEAN, MEDIAN, AND MODE

Component Overview: Students work independently or in pairs to interpret data in a table. Students find the mean and median of the data set and describe what they mean in the context of the question. Circulate as students work to complete this section and make note of any misconceptions to address either with individual students or as a class.

QUESTIONING

As an extension, consider allowing students to discuss how the answers might change if they asked people on the street or a more general location instead of those boarding a cruise ship.



9.4.3 Try It: Mean, Median, and Mode

- Nine adults boarding a cruise ship at Port Canaveral in Florida were randomly surveyed on the total number of vacations they have taken in the last five years. Their totals are summarized below.

Number of Vacations Taken								
5	4	3	9	1	4	0	6	3

- What does the value of the median represent in the context?
Half of the vacationers have taken more than the median number of vacations in the last five years and half have taken fewer.
- What is the median number of vacations taken by the adults surveyed?
4
- Determine the mean of the data. Then, interpret the mean in terms of the context by writing a sentence to summarize. Round to the hundredths.
The mean is 3.89 vacations. The people surveyed have taken an average of 3.89 vacations in the last five years.



9.4.4 Exploration: Understanding Mode and Range

Work with your partner to complete the following.

1. The
☐ mean
☐ median
☒ mode
☐ range
 is the value occurring most often in

a set of data, which can be used to describe trends in data.

2. While shopping over the last month, Shep collected the price for a jar of pickles from eight stores he visited.

Price for a Jar of Pickles			
Store A	\$1.66	Store E	\$1.99
Store B	\$2.99	Store F	\$1.99
Store C	\$1.49	Store G	\$1.50
Store D	\$1.99	Store H	\$1.99

- a. What does a jar of pickles cost at most stores?
\$1.99
- b. This value is the
☐ median of the data set.
☒ mode of the data set.
- c. What is the most Shep will spend on a jar of pickles?
\$2.99
- d. What is the least Shep will spend on a jar of pickles?
\$1.49
- e. The range of a data set describes how spread out the data is. Determine the range in price for a jar of pickles.
 $\$2.99 - \$1.49 = \$1.50$

3. Shep also began collecting the price of his favorite cereal to determine where he would get the best price.

Price for a Box of Cereal			
Store A	\$3.79	Store E	\$3.99
Store B	\$3.89	Store F	\$3.89
Store C	\$3.99	Store G	\$4.29
Store D	\$3.89	Store H	\$4.59

- a. The mode price of his favorite box of cereal is **\$3.89**.
- b. What is the most Shep will spend on a box of cereal?
\$4.59
- c. What is the least Shep will spend on a box of cereal?
\$3.79
- d. The range for the price of a box of cereal is **\$0.80**.
- e. The range describes the
☐ center
☒ variability
 of the data set.
4. With your partner, discuss whether the price for pickles or the price for cereal has more variability in pricing. Summarize the discussion.

The price for pickles has more variability as seen by pickles having the larger range of prices.



Measures of variability are numerical values that measure how much a data set varies from a central value.



The range is a **measure of variability**. By comparing the ranges for two or more data sets, the data set that has more variability can be determined by finding which has a larger **range**.

9.4.4 EXPLORATION: UNDERSTANDING MODE AND RANGE

Component Overview: In this Exploration, students will work with a partner to discover the purpose of the mode and range using the definitions from the 9.4.2 Guided Instruction.

DIFFERENTIATE INSTRUCTION

To provide an additional visual entry point, consider encouraging students to represent the data on a number line for questions 2 and 3. This will clearly display the spread of the data and allow students to visually see the range and mode.

TEACHER MOVES

Think-Pair-Share

Implement a Think-Pair-Share routine before students complete question 4. This will give students a chance to consider what it means to have more variability and formalize their responses with a partner before writing. Circulate as students discuss and bring the class together to address any misconceptions that may arise. Students may not connect that a larger range indicates more variability. This is an important misconception to address with the class.

9.4.5 COLLABORATION: ANALYZING NUMERICAL DATA

Component Overview: In the Collaboration, students will work with a partner to identify characteristics of mean, median, and mode and calculate each category using various data sets. Encourage both partners to show their work on each step to lessen the number of errors.

DIFFERENTIATE INSTRUCTION

This 9.4.5 Collaboration provides opportunities for differentiation through partner discussion and pairings. Consider how you want to group students. Those who are struggling may benefit from being paired with students who have demonstrated or are on their way to showing mastery. The discussion-based nature of this 9.4.5 Collaboration will foster mathematical discourse and thinking. Students may benefit from hearing how others reason through this section.

TEACHER MOVES

MA.K12.MTR.7.1: Mathematics in Real-World Contexts

To provide a project-based learning opportunity, consider allowing students to find the heights of boys and girls in their own class and see how their data compares to the sample class. Fostering the connection between their own class and the sample class will encourage engagement with the material and investment in the content.



9.4.5 Collaboration: Analyzing Numerical Data

Alice collected data on the height of sixth graders in her school. Below are two tables with heights, in inches (in.), of some of her male and female classmates.

Height of Boys (in.)	
58 in.	57 in.
57 in.	56 in.
55 in.	56 in.
57 in.	59 in.
58 in.	57 in.

Height of Girls (in.)	
55 in.	48 in.
55 in.	59 in.
48 in.	50 in.
58 in.	55 in.
50 in.	56 in.

- With your partner, determine who will be Partner A and who will be Partner B. Then, follow the directions.

Partner A: Use data from the height of ten boys to find the mean, median, and mode of the boys' heights in Alice's class. Then, write each value in the table.

Partner B: Use data from the height of ten girls to find the mean, median, and mode of the girls' heights in Alice's class. Then, write each value in the table.

- Share the values you calculated with your partner. Then, add the values your partner calculated into your table.

Write a statement in the table interpreting each mean and a statement interpreting each mode.

- Compare your statements with your partner. If necessary, resolve any differences.

Height of Boys (in.)		Height of Girls (in.)	
Mean of boys' heights	57 in.	Mean of girls' heights	53.4 in.
Median of boys' heights	57 in.	Median of girls' heights	55 in.
Mode of boys' heights	57 in.	Mode of girls' heights	55 in.
Interpretation of the mean height of boys: <i>The average height of the boys sampled is 57 inches.</i>		Interpretation of the mean height of girls: <i>The average height of the girls sampled is 53.4 inches.</i>	
Interpretation of the mode height of boys: <i>The most common height of boys sampled is 57 inches.</i>		Interpretation of the mode height of girls: <i>The most common height of girls sampled is 55 inches.</i>	

- With your partner, determine the range of boys' heights. Then, determine the range of girls' heights.
Boys: 4 inches
Girls: 11 inches
- Explain whether boys or girls in Alice's class have more variability in their heights.
The girls have more variability because there is a greater range in their heights.

**9.4.6 Wrap-Up: Reflection**

1. Read each of the “I can” statements below. Check the boxes that best explain your understanding.

Answers will vary.

- ☐ I can find and interpret the median, mode, and range of a given data set.
- ☐ I understand how to find the mean, median, and mode of a data set.
- ☐ I need support on how to find the mean, median, and mode of a data set.
- ☐ I understand how to find the range of a data set.
- ☐ I need support on how to find the range of a data set.
- ☐ I understand how to interpret measures of central tendency.
- ☐ I need support on how to interpret measures of central tendency.

9.4.6 WRAP-UP: REFLECTION

Component Overview: *In this Wrap-Up, students will reflect on their understanding of the lesson, what areas they excel in, and what areas they need more support and help with.*

DIFFERENTIATE INSTRUCTION

Consider using Google Forms to survey students to identify who needs the lesson to be retaught or reinforced.